ASSIGNMENT -1

CO2 – JAVA STRINGS AND COLLECTION FRAMEWORK

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COURSE CODE: CSA0920

COURSE NAME: PROGRAMMING IN JAVA

ASSIGNMENT SET: DIRECT-DESCRIPTIVE

Q1. String Immutability Demonstration

Strings in Java are immutable, meaning any operation that appears to modify a string actually creates a brand-new String object in memory rather than changing the original. Write a Java program that demonstrates this behavior by starting with a string, performing multiple concatenation operations on it in a loop, and using reference/identity checks (or printed object hash codes) to show that each concatenation results in a new object rather than an in-place modification of the original string.

Concepts: String immutability, object identity.

GitHub Repository Link: https://github.com/prabhath191006/Java

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2. PROBLEM UNDERSTANDING

This program demonstrates the concept of String immutability in Java. It performs multiple concatenation operations on a string and displays the object's identity hash code after each operation. The changing identity hash codes show that every concatenation creates a new String object instead of modifying the existing one.

3. APPROACH / DESIGN NOTES

- Initialized a String object with a sample value.
- Used a for loop to perform multiple concatenation operations.
- Displayed the string after each concatenation.

4. SOURCE CODE

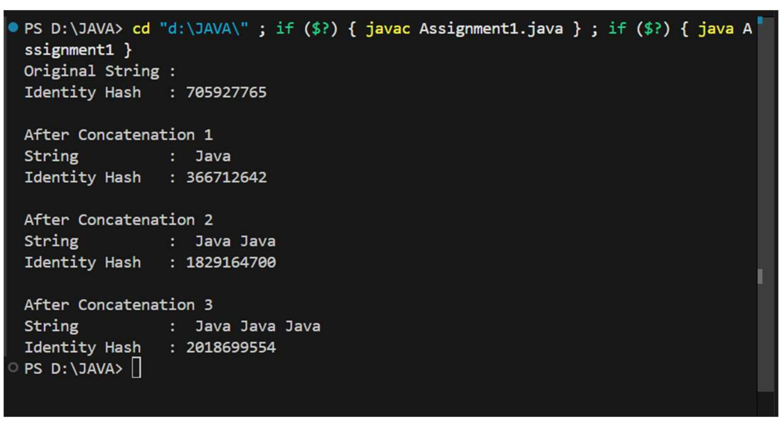
```
public class Assignment1 {

    public static void main(String[] args) {

        String str = "Hello";

        System.out.println("Original String : " + str);

        System.out.println("Identity Hash   : " + System.identityHashCode(str));

        for (int i = 1; i <= 3; i++) {

            str = str + " Java";

            System.out.println("\nAfter Concatenation " + i);

            System.out.println("String          : " + str);

            System.out.println("Identity Hash   : " + System.identityHashCode(str));

        }

    }

}
```

5. TEST CASES

#	Input	Expected Output	Actual Output
1	"Hello"	A new identity hash code is displayed after each concatenation.	Pass
2	"" (Empty String)	Concatenation creates a new String object with a different identity hash code.	Pass
3	Multiple concatenations	Every concatenation results in a new identity hash code.	Pass

6. SCREENSHOTS OF EXECUTION

TEST CASE - 1



TEST CASE - 2

```
PS D:\JAVA> cd "d:\JAVA\" ; if ($?) { javac Assignment1.java } ; if ($?) { java Assignment1 }
Original String : Hello
Identity Hash   : 705927765

After Concatenation 1
String          : Hello Java
Identity Hash   : 366712642

After Concatenation 2
String          : Hello Java Java
Identity Hash   : 1829164700

After Concatenation 3
String          : Hello Java Java Java
Identity Hash   : 2018699554
PS D:\JAVA>
```

TEST CASE - 3

```
PS D:\JAVA> cd "d:\JAVA\" ; if ($?) { javac Assignment1.java } ; if ($?) { java Assignment1 }
Original String : Java
Identity Hash   : 705927765

After Concatenation 1
String          : Java Java
Identity Hash   : 366712642

After Concatenation 2
String          : Java Java Java
Identity Hash   : 1829164700

After Concatenation 3
String          : Java Java Java Java
Identity Hash   : 2018699554
PS D:\JAVA>
```

7. REFLECTION

This assignment helped me understand that Java Strings are immutable and cannot be modified after creation. I learned that each concatenation operation creates a new String object, which can be verified using the object's identity hash code.

8. DECLARATION

This is my own work. Any AI/external tool assistance used is disclosed above.